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Urban-scale Prediction Scientific Software Engineer

Met Office

Apply before 11:55 pm on Sunday 1st November 2026

Reference number

486602

Salary

£35,145 - £37,701

Your total reward package will be up to £51,448 annually, which includes an RRA of £2,000 per annum as a market supplement to reflect the demand for your skills.

A Civil Service Pension with an employer contribution of 28.97%

Job grade

Higher Executive Officer

Contract type

Permanent

Type of role

Engineering

Working pattern

Flexible working, Full-time

Number of jobs available

1

Contents

Location
About the job
Benefits
Things you need to know
Apply and further information

Location

Reading, South East England, RG6 6ET

About the job

Job summary

We re looking for an exceptional **Urban-scale Prediction Scientific Software Engineer** to help us make a difference to our planet.

As our Urban-scale Prediction Scientific Software Engineer, the job may be suitable for hybrid working, which is where an employee works part of the week in the office and part of the week from home. This is a voluntary, non-contractual arrangement and the location advertised will be your contractual place of work.

Our opportunity is full time, 37 hours per week. Our people are at the heart of what we do and we'll do our best to agree a working pattern that works for everyone.

World changing work

From science to technology, from meteorology to management, and from planning to communication, our expertise helps us stand out as the authority on weather accuracy and climate prediction. We help individuals, industries and government to make better decisions to stay safe and thrive. This is the Met Office. This is who we are.

- We re a force for good - focusing on our environmental and social impact
- We re experts by nature - always learning and developing to do things better
- We live and breathe it - putting our purpose at the heart of decision-making
- We re better together - understanding partnerships and inclusivity make us greater
- We keep evolving - pushing boundaries to make tomorrow better for our customers

Your world of expertise

You will join a team developing 100m-scale urban prediction techniques to forecast small-scale hazards like thunderstorms, heat, and fog. High-

resolution physical models are computationally prohibitively expensive for routine use, creating a clear opportunity for data-driven solutions. Utilising our unique high-resolution training datasets, you will build, optimise, and implement machine learning models and workflows into practical, routine-running applications. Based at our University of Reading hub, you will apply advanced ML engineering to solve urban weather challenges alongside academic and research specialists.

Job description

Contributing to research and software delivery for urban-scale machine learning, including refactoring, modularizing, and optimizing ML workflows; evaluating and adopting emerging algorithms; implementing software best practices; and optimizing code performance across high-performance computing (HPC, including GPUs) and cloud computing platforms.

Working closely with colleagues across different teams within the Met Office to ensure consistency of approach and software standardisation.

Working collaboratively with partners in UK universities and UM partner organisations to advance partnerships on sub-km/urban prediction applications, noting in particular links with University of Reading scientists which are currently being strengthened through the new Local Hazard Prediction Centre of Excellence.

Documenting and presenting the work in internal reports, meetings, scientific journals, and external conferences. To provide scientific/ technical advice to peers, management and external stakeholders.

Person specification

1. **Lead Criterion:** Show technical insight through delivery of significant advances in software capability by the effective application of technical knowledge, understanding and skills (Expert By Nature)
2. An understanding of, demonstrable interest in, or experience of machine learning, AI, or data science methods to solve complex problems (Experts by nature)
3. Apply good Quality Assurance processes, best practice, standards and/or regulations in your work (Live and Breathe It)

4. Engage with others to understand the requirements for your work and its wider context, and contribute to the development of plans to meet these requirements (Better Together)
5. Communicate knowledge accurately and concisely in written documents and group discussions, tailoring communication for diverse audiences and building trust/credibility (Better Together)
6. Proactively develop your skills and support others in developing their skills, leading to beneficial impacts (Keep Evolving)

The panel may perform a preliminary sift of the lead criteria as indicated above.

Benefits

Alongside your salary of £35,145, Met Office contributes £10,181 towards you being a member of the Civil Service Defined Benefit Pension scheme. Find out what benefits a Civil Service Pension provides (opens in a new window).

Our work is life-changing, often life-saving and always life-enhancing. The Met Office is Great Place to Work UK certified. We are also featured on their Best Workplaces in Tech 2023, 2024 and 2025 lists, as well as their 54 Best Workplaces for Women 2023 list.

As our Urban-scale Prediction Scientific Software Engineer, your total reward package will be up to £51,448 annually, which includes:

£35,145 base pay

An outstanding Civil Service pension, with an average employer contribution of 28.97%

Recruitment Retention Allowance (RRA) you will be paid £2,000 per annum as a market supplement to reflect the demand for your skills. Whilst in post, you will be paid this market driven allowance from April 2026 until March 2027 in your monthly pay

Annual Leave starting at 27.5 days (plus Bank Holidays) rising to 32.5 days (plus Bank Holidays) after 5 years and option to buy or sell up to 5 days per year of annual leave

Things you need to know

Artificial intelligence

Artificial intelligence can be a useful tool to support your application, however, all examples and statements provided must be truthful, factually accurate and taken directly from your own experience. Where plagiarism has been identified (presenting the ideas and experiences of others, or generated by artificial intelligence, as your own) applications may be withdrawn and internal candidates may be subject to disciplinary action. Please see our candidate guidance (opens in a new window) for more information on appropriate and inappropriate use.

Selection process details

If you share our values, we d love to hear from you! Click apply to begin your application. Please complete your career history and provide evidence against each of the essential criteria in the supporting statement questionnaire. We recommend candidates use the CARL method (Context, Action, Result and Learning) for presenting evidence of experience and skills.

Please note: We are currently recruiting for 1 role which is being advertised at 2 levels (Foundation Scientific Software Engineer and Scientific Software Engineer). Those wishing to be considered at multiple levels should submit an application at each level.

Closing date 01/11/2026 at 23:59 with first stage interviews commencing from 16/11/2026. You will hear from us once the closing date has passed.

Using AI in your application

We welcome applications that use AI tools for support in drafting or refining, as long as they accurately reflect your own skills and experience. All hiring decisions at the Met Office are made by people, not AI. For more details, visit our approach to recruitment.

How we can help

If you have any questions or would like to discuss this opportunity further, please contact us at careers@metoffice.gov.uk.

If you re considering applying and need support to do so, please get in touch. You can request adjustments either within your application or by contacting us. Should you be offered an interview, please be aware there may be a selection exercise which could include a presentation, written test or a scenario-based activity. You can select in your application to be considered under the Disability Confident Scheme. To be invited to interview/assessment under this scheme, your application must meet the essential criteria for the role.

We understand that great minds don t always think alike and as an equal opportunities employer we welcome applications from those with all protected characteristics. We recruit on merit, fairness, and open competition in line with the Civil Service Code.

We can only accept applications from those eligible to live and work in the UK - please refer to GOV.UK for information. We require Security clearance without caveats and/or restrictions for this role, for which you need to have resided in the UK for at least 3 of the last 5 years to be eligible, 2 of these years must be immediately preceding the point of your application.

If you feel that your application has not been treated in accordance with the Recruitment Principles, and wish to make a complaint, then in the first instance you should contact us at: **careers@metoffice.gov.uk** If you are not satisfied with the response that you receive, then you can contact the Civil Service Commission at: **info@csc.gov.uk**

Feedback will only be provided if you attend an interview or assessment.

Security

Successful candidates must meet the security requirements before they can be appointed. The level of security needed is security check (opens in a new window).

See our vetting charter (opens in a new window).

People working with government assets must complete baseline personnel security standard (opens in new window) checks.

Medical

Successful candidates will be expected to have a medical.

Nationality requirements

This job is broadly open to the following groups:

- UK nationals
- nationals of the Republic of Ireland
- nationals of Commonwealth countries who have the right to work in the UK
- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities with settled or pre-settled status under the European Union Settlement Scheme (EUSS) (opens in a new window)
- nationals of the EU, Switzerland, Norway, Iceland or Liechtenstein and family members of those nationalities who have made a valid application for settled or pre-settled status under the European Union Settlement Scheme (EUSS)
- individuals with limited leave to remain or indefinite leave to remain who were eligible to apply for EUSS on or before 31 December 2020
- Turkish nationals, and certain family members of Turkish nationals, who have accrued the right to work in the Civil Service

Further information on nationality requirements (opens in a new window)

Working for the Civil Service

The Civil Service Code (opens in a new window) sets out the standards of behaviour expected of civil servants.

We recruit by merit on the basis of fair and open competition, as outlined in the Civil Service Commission's recruitment principles (opens in a new window).

The Civil Service embraces diversity and promotes equal opportunities. As such, we run a Disability Confident Scheme (DCS) for candidates with disabilities who meet the minimum selection criteria.

Diversity and Inclusion

The Civil Service is committed to attract, retain and invest in talent wherever it is found. To learn more please see the Civil Service People Plan (opens in a new window) and the Civil Service Diversity and Inclusion Strategy (opens in a new window).

Apply and further information

The Civil Service welcomes applications from people who have recently left prison or have an unspent conviction. Read more about prison leaver recruitment (opens in new window).

Once this job has closed, the job advert will no longer be available. You may want to save a copy for your records.

Contact point for applicants

Job contact :

Name : Resourcing Team

Email : careers@metoffice.gov.uk

Recruitment team

Email : careers@metoffice.gov.uk

Further information

careers@metoffice.gov.uk

Civil Service Commission

disability confident EMPLOYER